

CMSE 801 - Day 9

Compartmental Modeling

Oct 3, 2022

General Course Announcements

- Homework 2 has been released Tuesday Sep 27 and is due Wednesday Oct 12 at 11:59 p.m.
- Midterm will be on Wednesday Oct 19 during the class
- The details of the semester project have been released

Questions/comments from Day 9 PCA

- I don't understand how the guide for the 3D plot gave the format "ax = fig.add_subplot(111, projection='3d')". How can I utilize this in a subplot given the structure we have covered?
- we never talked about what a tuple is so that was confusing.
- Could you explain more about this assignment during class? Especially defining the function and the 3D plot. Thanks.
- More clarification about compartmental models, more examples, and just clarification about how it can be applied to programming.
- Going over ODEs in general, like how do we come up with the equations that were given to us here?
- I would like to know what was wrong with my function
- What is the difference between plot.surface and plot.trisurf, I mean what does it mean to take x,y, and z as 1D arrays or 2D arrays?
- How to create 3D plot by using subplot function?
- how do we deal with silent errors?
- Yes, when I do clear all and rerun on the kernel my plots work but then when I go to run the individual chunks of code where my plots are I get weird error messages.
- For 3D plot, I wonder how to rotate the angle of view. Mine is not in a satisfying angle so far.

```
In [1]: import matplotlib.pyplot as plt
        from mpl_toolkits.mplot3d import Axes3D
        import numpy as np
        %matplotlib inline

        # Note that the SciPy library is needed for odeint
        from scipy.integrate import odeint
```

```
In [2]: def derivs(y, t, beta, delta, p, r):

        # unpack list of incoming variables
        target = y[0]
```

```

infected = y[1]
virion = y[2]

dTdt = -beta*target*virion
didt = (beta*target*virion) - (delta*infected)
dVdt = (p*infected) - (r*virion)

return [dTdt, didt, dVdt]

```

```

In [3]: target_0 = 4e8
infected_0 = 0.0
virions_0 = 1.0
# Here we put them all into a list so that they can be passed to odeint
y0 = [target_0, infected_0, virions_0]

```

```

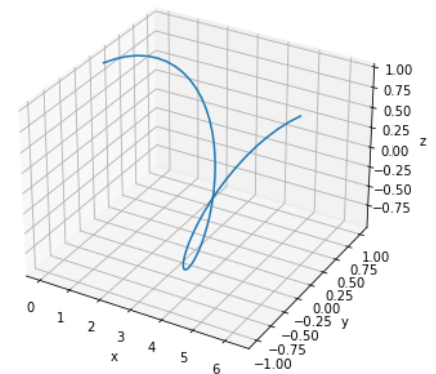
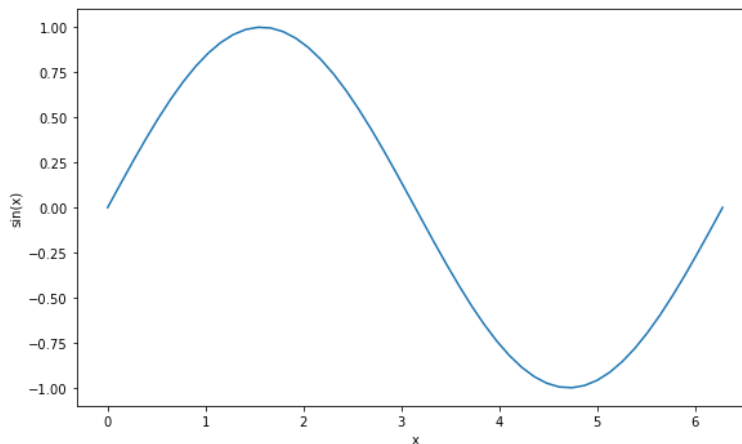
In [4]: x = np.linspace(0,2*np.pi,50)
y = np.sin(x)
z = np.cos(x)

```

```

In [5]: fig = plt.figure(figsize=(15,5))
ax1 = fig.add_subplot(1,2,1)
ax1.plot(x, y)
ax1.set_xlabel('x')
ax1.set_ylabel('sin(x)')
ax2 = fig.add_subplot(1,2,2, projection = '3d')
ax2.plot(x, y, z)
ax2.set_xlabel('x')
ax2.set_ylabel('y')
ax2.set_zlabel('z')
plt.tight_layout()

```



```

In [6]: fig = plt.figure(figsize=(15,5))
ax = fig.add_subplot(projection = '3d')
ax.plot(x, y, z)
ax.view_init(10,20)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('z')
plt.tight_layout()

```

